

# Prakhar Shekhar Parthasarathi

Full-Stack Software Engineer / Java & Spring Boot Backend / React Front-End / AWS

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🐙 github.com/praxstack · 🌐 linkedin.com/in/prakharshekhar ·

🌐 Portfolio · prax-portfolio-one.vercel.app

47%

P90 LATENCY CUT ·  
BOOKING SURFACES

+33%

CONVERSION LIFT

195k rps

GET · JAVA REDIS FROM  
SCRATCH

20→5

ON-CALL TICKETS /  
SHIFT

## ≡ SUMMARY

Full-stack software engineer with **3+ years at Amazon** across Payments and Travel — **owned 4 microservices end-to-end** through design, build, deploy, and on-call. Java-heavy backend (**Java 17 / Spring Boot**, JVM concurrency with `CompletableFuture`, bounded `ExecutorService`, `ConcurrentHashMap`), idempotent event-driven architecture over **SQS + Step Functions**, and typed infrastructure in **AWS CDK (Java)** — paired with **React (SSR & CSR) + Redux** for customer-facing flows. Customer-facing impact: **47% p90** reduction on booking surfaces, **33% conversion** lift, **43 bps CPT** reduction via idempotent schedule-change automation, on-call tickets cut **20+ to ~5** per shift.

## ⌘ CORE TECHNICAL STACK

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|  <b>JAVA &amp; JVM</b><br><b>Java 17</b> , Spring Boot, Spring MVC, Spring DI, Google Guice, JUnit 5, Mockito, SLF4J, Lombok, Maven, Gradle · <code>ExecutorService</code> , <code>CompletableFuture</code> , <code>ConcurrentHashMap</code> , thread-pool tuning, JVM profiling, GC analysis |  <b>DISTRIBUTED SYSTEMS</b><br>Microservices, REST, GraphQL, event-driven architecture, idempotency & dedup keys, state machines, retry with exponential backoff, DLQ handling, at-least-once semantics, TDD |
|  <b>AWS</b><br>Lambda (Java), DynamoDB, SQS, Step Functions, API Gateway, AppSync, CloudWatch, ECS, <b>CDK (Java)</b> — typed IaC for all service infrastructure                                                                                                                              |  <b>FRONT-END</b><br><b>React.js (SSR &amp; CSR)</b> , Redux (1.5 yrs production) — customer-facing Amazon flows; HTML / CSS, JavaScript / TypeScript                                                        |
|  <b>DATA &amp; STORAGE</b><br>DynamoDB, Redis, NoSQL modelling, SQL, caching strategies                                                                                                                                                                                                       |  <b>OTHER LANGUAGES</b><br>Python, C++, VTL                                                                                                                                                                  |
|  <b>TOOLING &amp; PROCESS</b><br>Git, GitHub, CI/CD, Gradle JaCoCo coverage gates, Agile / Scrum, on-call, code review, incident response                                                                                                                                                     |                                                                                                                                                                                                                                                                                                 |

## Software Development Engineer

Jul 2022 – Sep 2025

📍 Amazon · Travel (Hotels & Flights) · Bengaluru

- **Service ownership — Spring Boot, Spring MVC.** Owned 4 **customer-facing microservices** powering the hotel booking platform — search, listing, room selection, order review — across design, code review, deploy, and on-call. Built REST controllers, service-layer logic, and `@Repository` DynamoDB DAOs wired through Spring DI. Shipped Prime-exclusive discount flow and partner-discount integration with tuned cache logic — **~30 bps** drop in funnel churn.
- **JVM concurrency — CompletableFuture, bounded executors.** Refactored sequential downstream calls into `CompletableFuture.allOf()` on a bounded `ExecutorService`; parallelised 6 partner-API lookups — hotel landing p90 cut **47%** (**1.1s → 0.59s**), conversion up **+33%**. GZIP on 150–200 MB catalog payloads cut wire size **95%**, eliminating network-timeout failures.
- **Event-driven — SQS + Step Functions.** Designed idempotent Java webhook for real-time flight schedule changes (cancellations, date-changes, non-operational flights) — SQS ingestion, DynamoDB-backed dedup keys, audit logging, automated offline modifications — addressing the highest-impact customer-pain category (**43 bps CPT**). Orchestrated end-to-end booking state machine (confirm / cancel / retry) on Step Functions with Java Lambda handlers.
- **Infrastructure as code — AWS CDK (Java).** Authored all service IaC in **Java CDK** — Lambdas, DynamoDB, SQS, Step Functions, API Gateway, CloudWatch alarms. Migrated 2 services from hand-rolled CloudFormation to typed CDK stacks — compile-time infrastructure checks, **halved** rollback incidents.
- **Testing — JUnit 5, Mockito, Spring Boot.** Held **80%+** line coverage; Mockito argument captors for downstream contract verification; `@SpringBootTest` integration tests against in-memory DynamoDB. Introduced Gradle JaCoCo coverage gates in CI.
- **Operational excellence & high-bar automation.** Dove deep into multi-year on-call corpus — tickets cut from **20+ to ~5 per shift** (**~2 SDE-weeks / week** reclaimed). Java email-automation service deprecated stale WebLab experiments — config-hygiene incidents from 5–6 to **0** per on-call. Resolved **150+** production incidents; authored COEs and drove action items to closure. Built auto-insurance purchase flow (Java + React) with milestone persistence and auto-fill — **60%+** lift in renewals.

## Software Development Engineer, Intern

May 2021 – Jul 2021 · Feb 2022 – Jul 2022

📍 Amazon · Bengaluru

- Migrated legacy GraphQL resolvers to **Java** — rebuilt resolver classes, wired them to DynamoDB via Spring-injected DAOs, exposed through API Gateway + Lambda.
- Shipped **One-Click Renewal** for auto-insurance using past-purchase data — async Java pre-fill workflow that materially simplified the funnel.
- Applied OO design and **Google Guice DI** throughout; landed a clean, testable codebase with **90%+** unit-test coverage (JUnit 5 + Mockito).

## 📁 SELECTED PROJECTS

### AI Visual Code Review

[github.com/praxstack/ai-visual-code-review](https://github.com/praxstack/ai-visual-code-review) · npm

Node.js Express TypeScript Jest VS Code API

Security-hardened visual code-review tool with a GitHub-style diff interface, real-time Git status tracking, AI-optimized markdown exports for ChatGPT / Claude, command-injection protection, and rate limiting. Published to npm and shipped as a VS Code extension.

Published: npm v2.4.1 · MIT · CLI + VS Code extension.

### Redis Server in Java

[github.com/praxstack/redis-server-java](https://github.com/praxstack/redis-server-java)

Java 17 Maven JUnit 5 Mockito  
ConcurrentHashMap ScheduledExecutorService

RESP2-compatible Redis server built from scratch. Thread-per-client on a bounded `ExecutorService`; lock-free reads via `ConcurrentHashMap`; hybrid TTL eviction (lazy check on `GET` plus active `ScheduledExecutor` sweep); graceful shutdown via `Runtime.addShutdownHook`; integration tests over real TCP sockets.

**Benchmarks** (redis-benchmark, 100 concurrent clients):  
SET **154k rps** · GET **195k rps** · INCR **128k rps**  
· p50 < **0.3 ms** · **42 tests, 100% pass.**

### Audio Transcription Pipeline

[github.com/praxstack/audio-transcription-pipeline](https://github.com/praxstack/audio-transcription-pipeline)

Python AWS Bedrock Voxtral Claude pyannote  
pytest

Multi-stage pipeline turning raw audio/video into clean, speaker-labelled, optionally-translated Markdown. Pairs local pyannote diarization with AWS Bedrock (Voxtral ASR + Claude script-repair) across deterministic, resumable file-in/file-out stages — domain-neutral and code-switch aware.

**Metrics:** **63** tests · 6 pipeline stages · local diarization · MIT.

### Coach Atlas

[github.com/praxstack/coach-atlas](https://github.com/praxstack/coach-atlas)

React TypeScript Vite Bedrock Mermaid  
KaTeX

An AI technical-interview mentor and content studio — Socratic interview prep, production-grade tutorial generation, and Mermaid / KaTeX / code rendering. Multi-provider (OpenAI, Anthropic, Gemini, AWS Bedrock) with a BYOK design: keys live in the browser, never on a server.

**Stack:** React + TypeScript · multi-provider BYOK · deployed on Vercel.

## Open-Source Contributions

Merged into [danielmiessler/Fabric](#) (Go, 42k★ AI-tooling framework) — three production PRs:

**#2044** — added AWS Bedrock bearer-token (ABSK) authentication + guided `--setup` (the same auth Claude Code uses).  
**+882 / -83**

**#2052** — dynamic Bedrock region fetching from botocore (40+ regions) + AWS\_PROFILE conflict fix. **+252 / -15**

**#2061** — fixed a streaming-goroutine deadlock in `Chatter.Send` + unified CLI/REST strategy handling. **+494 / -42**

Open PRs under review: [bytedance/deer-flow](#) (75k★ · model-config normalization), [thedotmack/claude-mem](#) (85k★ · plugin packaging), [refactoringhq/tolaria](#) (17k★ · editor callouts).

## 🎓 EDUCATION

**M.C.A., Computer Science** · Motilal Nehru National Institute of Technology (MNNIT), Prayagraj **6PA** **2019 – 2022**  
**9.25**

**B.Sc. (Hons), Computer Science** · Bhaskaracharya College of Applied Sciences, University of Delhi **2015 – 2018**

## 🏆 AWARDS & RECOGNITION

**🏆 Amazon Pay Merchants Categories Champion** — 100+ hotel trust-buster bugs resolved; on-call tickets **2023**  
cut **20+ to ~5** per shift.